

SPRiF: Spectral Projective Reset Integrate-and-Fire Neuron

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Abstract

Temporal tasks often require information to persist across long intervals. Standard spiking neurons use the same membrane state to accumulate context, generate spikes, and receive post-spike reset. This coupling repeatedly perturbs the temporal trace that a neuron must preserve. We introduce **SPRiF** (Spectral Projective Reset Integrate-and-Fire Neuron), a spiking unit based on *functional state decomposition*. SPRiF separates a constrained spectral slow state, which provides real and rotational memory modes without reset, from a fast discharge state that generates spikes and receives a learnable projective reset. The design preserves a resettable spike-generation pathway while insulating slow temporal memory from post-spike correction. Across five temporal classification benchmarks, SPRiF attains the highest mean accuracy among the compared methods on three tasks and remains competitive on the other two. Ablations identify slow-fast state separation as the primary contributor, with consistent gains from spectral rotation and projective reset. A controlled spike-intervention task shows stable delay-period recall under reset stress, and impulse-response analysis reveals heterogeneous temporal kernels. These results support functional state decomposition as a practical neuron-level design principle for temporal SNNs.

1 Introduction

Spiking neural networks (SNNs) process temporal information through discrete spike events. They are used in speech recognition, physiological signal analysis, and event-based recognition. Unlike continuous-activation networks, their representational capacity depends on both inter-layer connectivity and neuron dynamics. These dynamics determine how neurons accumulate temporal context, decide when to fire, and update their states after each spike. When relevant information spans multiple timescales, from speech phonemes to long permuted sequences, neuron-level dynamics constrain the temporal patterns that an SNN can retain and exploit.

The leaky integrate-and-fire (LIF) neuron remains the dominant SNN unit because it is simple and compatible with surrogate-gradient training. Its extensions enrich neuronal

dynamics in several ways. Adaptive-threshold models introduce slow variables that modulate firing (Baronig et al. 2025; Zhang et al. 2025b; Idrees et al. 2024). Resonant neurons add damped subthreshold oscillations (Higuchi et al. 2024; Huang et al. 2024b; Zhang et al. 2025a), and multi-timescale designs use heterogeneous neuronal or synaptic decays (Wei et al. 2026; Deng et al. 2025; Fullea-Garcia, Soldado-Magraner, and Margarit-Taulé 2026; Tiwari and Chauhan 2025). Recent reset mechanisms also modify the post-spike update to reduce information loss (Deng et al. 2022; Huang et al. 2025; Li et al. 2025; Fang et al. 2023). Despite these advances, the spike-generating variable is usually also the reset target. In standard LIF, one membrane potential accumulates history, determines spike timing, and absorbs reset. Each spike therefore perturbs its temporal trace. Oscillatory and adaptive variants retain this membrane-centred organisation, whereas reset innovations soften, adapt, or remove reset from the same spike-generating state.

This coupling raises a basic question: **must the state that triggers a spike also store temporal memory?** A spike indicates that a discharge-related variable has crossed a threshold and therefore requires correction. It need not alter every accumulated memory trace. Insulating a memory-bearing state from spike-triggered reset could preserve a richer input history without abandoning binary communication or enlarging the network.

We address this question with **SPRiF** (Spectral Projective Reset Integrate-and-Fire Neuron), which implements *functional state decomposition*. Each neuron contains a slow spectral state and a fast discharge state. The slow state integrates input through one real exponential mode and one two-dimensional damped rotational mode. It is never reset by spikes. The fast state receives a projection of the slow state, produces the membrane readout and binary spikes, and absorbs a learnable *projective reset* along $[1, \lambda]^T$. SPRiF is trained using standard backpropagation through time and surrogate gradients. Spike events therefore correct only the fast discharge state while the slow state evolves continuously.

Functional decomposition differs from conventional multi-timescale design. The latter assigns different decay rates to parallel channels, a form of *rate decomposition*, while coupling their roles to readout and reset. SPRiF instead assigns distinct functions. The constrained slow state stores temporal context, and the fast state implements readout and

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reset. The reset operator has no update path to the slow state. The distinction is therefore structural and testable at the update level, rather than a consequence of heterogeneous time constants alone.

We evaluate SPRiF on five temporal benchmarks spanning sequential image classification, physiological signal analysis, speech recognition, and event-based digit recognition. SPRiF attains the highest compared mean accuracy on S-MNIST, PS-MNIST, and QTDB. It remains close to the strongest compared results on GSC and SHD. Its end-to-end parameter count is lower than that of the strongest compared entry on four datasets and comparable on QTDB. Ablations on PS-MNIST, GSC, and QTDB isolate the contributions of rotation, state separation, and projective reset. A controlled spike-intervention experiment and impulse-response analyses further examine the proposed dynamics.

Contributions.

1. We propose *functional state decomposition* for spiking neurons, separating temporal memory storage, spike read-out, and post-spike reset into distinct state components. This principle is orthogonal to rate decomposition and single-state reset.
2. We introduce a *constrained spectral slow state* with real exponential decay and damped rotation modes, providing interpretable, structured temporal filtering beyond the single-exponential LIF trace.
3. We propose *projective fast-state reset*, a learnable directional reset mechanism that operates exclusively on the discharge state and leaves the memory state untouched.
4. We evaluate SPRiF on five temporal benchmarks, three mechanism ablations, a controlled spike-intervention experiment, and analyses of learned temporal kernels. Code will be released upon acceptance.

2 Related Work

2.1 Neuron Dynamics for Temporal Processing

Extensions of the leaky integrate-and-fire (LIF) neuron enrich temporal dynamics through gated integration, conductance-based modelling, adaptive thresholds, and improved surrogate-gradient training (Fang et al. 2022; Bai, Wang, and Yu 2026; Yao et al. 2021; Huang et al. 2024a; Kamata et al. 2022). Adaptive and multi-timescale models introduce slow variables, heterogeneous neuronal or synaptic decays, conductance channels, and learnable temporal gates (Baronig et al. 2025; Zhang et al. 2025b; Idrees et al. 2024; Wei et al. 2026; Deng et al. 2025; Fullea-Garcia, Soldado-Magraner, and Margarit-Taulé 2026; Tiwari and Chauhan 2025). These approaches provide *rate decomposition*: parallel components evolve at different timescales, but remain coupled to membrane readout and reset.

Oscillatory and dendritic neurons offer complementary routes to temporal selectivity. Resonate-and-fire variants use damped oscillations (Izhikevich 2001; Higuchi et al. 2024; Huang et al. 2024b; Zhang et al. 2025a), whereas compartmental and heterogeneous models introduce dendritic structure and dynamics (Yin et al. 2024; Zheng et al. 2024; Stevenson and Kappel 2023; Lange et al. 2021). Structured

state-space models have also been incorporated into spiking neurons (Fabre 2025; Shen et al. 2025). SPRiF differs from these alternatives along three axes. It assigns memory and discharge to distinct state roles, confines reset to the discharge state, and constrains memory dynamics to one real and one rotational spectral mode. Its contribution is therefore a structurally separated memory–reset interface, rather than heterogeneous rates or spatial compartments alone.

2.2 Reset and Functional State Separation

Existing reset mechanisms soften, adapt, or remove the post-spike update to reduce information loss and, in some cases, enable parallel training (Deng et al. 2022; Li et al. 2025; Huang et al. 2025; Zhang 2025; Fang et al. 2023). They still modify the spike-generating state itself, or remove reset from that state. SPRiF instead confines a learnable projective reset to the fast discharge state and leaves the slow spectral state untouched. Network-level temporal architectures, including temporal subnetworks and stochastic spike communication (Ding et al. 2025b, 2024, 2025a), are complementary to this neuron-level separation.

3 Method

3.1 From Conventional LIF Dynamics to Functional Decomposition

We begin with the discrete-time dynamics of a standard leaky integrate-and-fire (LIF) neuron. Given an input current I_t , a scalar membrane state integrates its history and then generates a binary spike,

$$\tilde{v}_t = \beta v_{t-1} + I_t, \quad z_t = H(\tilde{v}_t - \theta), \quad (1)$$

followed by the post-spike update

$$v_t = \tilde{v}_t - z_t \theta, \quad (2)$$

where $\beta \in (0, 1)$ is the leak factor, H is the Heaviside step function, and θ is the firing threshold. This formulation is computationally attractive, but assigns three functions to the same state. The state v_t retains temporal context, determines spike timing, and receives reset. Every spike therefore corrects the state that also carries neuronal memory. Without reset, the resulting memory kernel is a single exponential, proportional to β^k after k steps.

SPRiF retains the event-driven LIF interface but separates these functions. For a layer input $\mathbf{s}_t$, the current is

$$I_t = W_{\text{in}} \mathbf{s}_t + W_{\text{rec}} \mathbf{z}_{t-1} + b, \quad (3)$$

where the recurrent term is omitted in feed-forward layers. Rather than using I_t to update one resettable membrane variable, each SPRiF neuron maintains a three-dimensional slow state $\mathbf{x}_t \in \mathbb{R}^3$ and a two-dimensional fast state $\mathbf{u}_t \in \mathbb{R}^2$. The slow state carries temporal context. The fast state converts that context into a membrane readout, spikes, and post-spike correction. The ordered computation is

$$I_t \rightarrow \mathbf{x}_t \rightarrow \tilde{\mathbf{u}}_t \rightarrow v_t \rightarrow z_t \rightarrow \mathbf{u}_t, \quad (4)$$

where $\tilde{\mathbf{u}}_t$ denotes the pre-reset fast state. The two paths meet through a learned slow-to-fast projection, but only the fast path is reset. The following subsections specify the constrained slow memory system and the resettable discharge system.

3.2 Constrained Spectral Slow State

The slow state must retain context without direct spike-triggered correction and represent a richer temporal basis than the single exponential of Eq. (1). We therefore use the smallest real-valued linear state containing a real decay mode and a complex-conjugate rotational pair. Its update is

$$\mathbf{x}_t = A\mathbf{x}_{t-1} + BI_t, \quad (5)$$

with

$$A = \begin{bmatrix} \alpha & 0 & 0 \\ 0 & \rho \cos \omega & -\rho \sin \omega \\ 0 & \rho \sin \omega & \rho \cos \omega \end{bmatrix}, \quad B = \begin{bmatrix} 1 - \alpha \\ 1 - \rho \\ 0 \end{bmatrix}. \quad (6)$$

The spectrum of A is

$$\text{eig}(A) = \{\alpha, \rho e^{i\omega}, \rho e^{-i\omega}\}, \quad (7)$$

so the first coordinate is an exponential trace, $x_t^0 = \alpha x_{t-1}^0 + (1 - \alpha)I_t$, while (x_t^1, x_t^2) form a damped rotation driven through the first oscillatory coordinate. We parameterize

$$\alpha = \sigma(\alpha_{\text{raw}}), \quad \rho = \sigma(\rho_{\text{raw}}), \quad \omega = \pi\sigma(\omega_{\text{raw}}), \quad (8)$$

where σ is the logistic sigmoid, so that $\alpha, \rho \in (0, 1)$ and $\omega \in (0, \pi)$. These constraints keep all slow modes stable and retain a direct dynamical interpretation. The parameter α controls real decay, ρ controls rotational damping, and ω controls rotational frequency. They also prevent the memory transition from becoming an unconstrained recurrent matrix.

The structure makes the temporal basis explicit. After k steps, an input impulse contributes a term proportional to $(1 - \alpha)\alpha^k$ through the real mode. It contributes terms proportional to $(1 - \rho)\rho^k \cos(k\omega)$ and $(1 - \rho)\rho^k \sin(k\omega)$ through the rotational pair. SPRiF can therefore combine monotone and oscillatory traces within each neuron while retaining a linear-time recurrence. This state is computed before spike generation and is absent from the reset equation introduced next.

3.3 Fast Discharge State and Projective Reset

The slow state is mapped to a two-dimensional pre-reset fast state:

$$\tilde{u}_t^0 = \eta_0 u_{t-1}^0 + \kappa u_{t-1}^1 + (G\mathbf{x}_t)^0, \quad (9)$$

$$\tilde{u}_t^1 = \eta_1 u_{t-1}^1 + (G\mathbf{x}_t)^1, \quad (10)$$

where $\eta_0, \eta_1 \in (0, 1)$ are learned leak coefficients, κ couples the second fast coordinate to the first, and $G \in \mathbb{R}^{2 \times 3}$ projects slow dynamics into the discharge space. The first coordinate is the membrane potential, $v_t = \tilde{u}_t^0$. The forward spike is binary:

$$z_t = H(v_t - \theta). \quad (11)$$

where H is the Heaviside step function and θ is the firing threshold. We train through the discontinuity with the multi-Gaussian surrogate used in recurrent SNN training (Yao et al. 2021). The forward pass always uses the hard threshold.

After a spike, SPRiF corrects only the fast state:

$$\mathbf{u}_t = \tilde{\mathbf{u}}_t - z_t \theta \begin{bmatrix} 1 \\ \lambda_j \end{bmatrix}, \quad (12)$$

where λ_j is learned independently for neuron j . Setting $\lambda_j = 0$ recovers scalar subtraction on the membrane coordinate. Nonzero values allow reset to modify both discharge coordinates along a learned direction. Crucially, Eq. (12) contains no $\mathbf{x}_t$ term. A spike changes $\mathbf{u}_t$ but leaves the slow memory state intact. This structure preserves memory across spikes without removing reset or replacing binary communication.

3.4 Layer Construction, Training, and Cost

A SPRiF layer applies Eqs. (3)–(12) independently to each neuron and emits $[\mathbf{z}_1, \dots, \mathbf{z}_T]$. Layers can be stacked, and recurrence is optional rather than intrinsic to the neuron definition. Sequence-level tasks use a linear readout over temporally aggregated hidden spikes. QTDB uses a time-distributed classifier, and SHD uses a spiking readout whose membrane evidence is accumulated after a short warm-up. All parameters, including $\alpha, \rho, \omega, G, \kappa, \eta$, and λ , are jointly optimized with network weights through backpropagation in time.

Relative to a scalar LIF neuron, SPRiF stores five state values and adds 13 neuron-wise dynamics parameters: three spectral parameters, two fast leaks, one coupling, six slow-to-fast projection coefficients, and one reset coefficient. The structured state updates remain $O(h)$ for a layer of width h . When recurrent connectivity is enabled, the shared $O(h^2)$ matrix multiplication dominates both LIF and SPRiF variants. We report end-to-end parameter counts to quantify this trade-off.

4 Experiments

4.1 Benchmarks and Protocol

We tested whether a single neuron design transfers across five temporal regimes. S-MNIST and PS-MNIST present 784 normalized pixels sequentially, with PS-MNIST using a fixed permutation. QTDB is a six-class ECG sequence-classification task. GSC uses 101-frame, 120-dimensional mel-spectrograms for 12-class keyword spotting. SHD contains 700-channel event sequences from 20 spoken-digit classes. Accuracy is the evaluation metric throughout.

The S-MNIST and PS-MNIST networks used two recurrent SPRiF layers of widths 64 and 256. They were trained for 150 epochs with Adam, a batch size of 512, a learning rate of 10^{-2} , and truncated BPTT chunks of 262 steps. QTDB used one recurrent layer of width 36, Adam, a batch size of 128, a learning rate of 10^{-2} , and 250 epochs. GSC used one recurrent layer of width 300, temporal mean pooling, AdamW, a batch size of 128, a learning rate of 5×10^{-3} , and at most 150 epochs. SHD used a recurrent hidden layer of width 64 and a 20-neuron SPRiF readout, Adam, a batch size of 100, a learning rate of 10^{-2} , and at most 100 epochs. Spectral parameters were initialized from log-uniform time-constant ranges and learned without clipping under the constrained parameterization. Full task-specific settings, including seed lists, are provided in the technical appendix.

We compared SPRiF with published recurrent SNNs spanning gated, adaptive, dendritic, resonate-and-fire, and recent

Table 1: Comparison across five temporal benchmarks. “Rec” denotes a recurrent network, and parameters are reported in millions (M). SPRiF entries (bold) are mean $\pm$ std over 5 seeds.

Datasets	Method	Network	Parameters (M)	Accuracy (%)
S-MNIST	GLIF _{NeurIPS, 2022} (Fang et al. 2022)	Rec	0.15	96.64
	ASRNN _{Nat. Mach. Intell., 2021} (Yao et al. 2021)	Rec	0.15	98.70
	DH-SNN _{Nat. Commun., 2024} (Zheng et al. 2024)	Rec	0.08	98.90
	BRF _{ICLR, 2024} (Higuchi et al. 2024)	Rec	0.069	99.10
	TC-LIF _{AAAI, 2024} (Yin et al. 2024)	Rec	0.15	99.20
	SPRiF (ours)	Rec	0.067	99.28 $\pm$ 0.06
PS-MNIST	GLIF _{NeurIPS, 2022} (Fang et al. 2022)	Rec	0.15	90.47
	ASRNN _{Nat. Mach. Intell., 2021} (Yao et al. 2021)	Rec	0.15	94.30
	DH-SNN _{Nat. Commun., 2024} (Zheng et al. 2024)	Rec	0.08	94.52
	BRF _{ICLR, 2024} (Higuchi et al. 2024)	Rec	0.069	95.20
	TC-LIF _{AAAI, 2024} (Yin et al. 2024)	Rec	0.15	95.36
	SPRiF (ours)	Rec	0.067	95.86 $\pm$ 0.14
QTDB	ASRNN _{Nat. Mach. Intell., 2021} (Yao et al. 2021)	Rec	0.0018	85.90
	DH-SNN _{Nat. Commun., 2024} (Zheng et al. 2024)	Rec	0.00178	86.35
	BRF _{ICLR, 2024} (Higuchi et al. 2024)	Rec	0.00173	87.00
	SE-adLIF _{Nat. Commun., 2025} (Baronig et al. 2025)	Rec	0.0018	86.88
	SPRiF (ours)	Rec	0.00177	88.43 $\pm$ 0.36
GSC	SNN-SFA _{eLife, 2021} (Salaj et al. 2021)	Rec	4.19	91.21
	ASRNN _{Nat. Mach. Intell., 2021} (Yao et al. 2021)	Rec	0.31	92.10
	RadLIF _{Front. Neurosci., 2022} (Kamata et al. 2022)	Rec	1.20	94.51
	DH-SNN _{Nat. Commun., 2024} (Zheng et al. 2024)	Rec	0.13	93.80
	TC-LIF _{AAAI, 2024} (Yin et al. 2024)	Rec	0.20	94.84
	SPRiF (ours)	Rec	0.13	94.55 $\pm$ 0.31
SHD	Heterogeneous SNN _{Nat. Commun., 2021} (Lange et al. 2021)	Rec	0.11	82.50
	ASRNN _{Nat. Mach. Intell., 2021} (Yao et al. 2021)	Rec	0.14	90.40
	DH-SNN _{Nat. Commun., 2024} (Zheng et al. 2024)	Rec	0.05	91.34
	BRF _{ICLR, 2024} (Higuchi et al. 2024)	Rec	0.11	91.70
	TC-LIF _{AAAI, 2024} (Yin et al. 2024)	Rec	0.14	88.91
	MPS-SNN _{ICLR, 2025} (Ding et al. 2025b)	Rec	0.39	91.19
	DGN _{ICLR, 2026} (Bai, Wang, and Yu 2026)	Rec	0.22	87.78
	SPRiF (ours)	Rec	0.05	91.52 $\pm$ 0.46

temporal architectures. Source studies used task-specific protocols. Table 1 therefore provides descriptive cross-paper comparisons, not significance claims. SPRiF entries report mean $\pm$ std over five seeds.

4.2 Main Results

Table 1 reports the main benchmark results. SPRiF attained the highest mean accuracy among the compared results on S-MNIST, PS-MNIST, and QTDB, with five-seed results of $99.28 \pm 0.06\%$, $95.86 \pm 0.14\%$, and $88.43 \pm 0.36\%$, respectively. The margins over the strongest available published entries were 0.08, 0.50, and 1.43 points. Because these entries used task-specific protocols, the margins are descriptive rather than cross-paper significance claims. On GSC and SHD, SPRiF reached $94.55 \pm 0.31\%$ and $91.52 \pm 0.46\%$, 0.29 and 0.18 points below the strongest compared result. Together, these comparisons show that the same neuron mechanism remained competitive across pixel sequences, physiological signals, frame-based speech, and event streams.

The comparison also indicates parameter efficiency. SPRiF used 0.067M parameters on S-MNIST and PS-MNIST, 1.77k on QTDB, 0.13M on GSC, and 0.05M on SHD. Relative to the strongest-accuracy entry for each

dataset, this corresponded to approximately 55% fewer parameters on both MNIST variants, a comparable count on QTDB (1.77k versus 1.73k), 35% fewer on GSC, and 55% fewer on SHD. The multi-state neuron therefore did not require a larger end-to-end network to attain the reported accuracies.

4.3 Mechanism Ablations

We isolated three design choices on PS-MNIST, GSC, and QTDB. Ablation A set $\omega = 0$, removing rotational coupling while retaining the real decay path. Ablation B merged memory and discharge into a single three-dimensional state, whose first coordinate served as both membrane and reset target. Ablation C fixed $\lambda = 0$, reducing Eq. (12) to scalar reset. All variants otherwise retained the task-specific training configuration.

Table 2 shows two consistent patterns. Merging the states reduced accuracy by 2.38–5.04 points. Thus, the benefit of state separation extends beyond increased state dimensionality: the same three-dimensional capacity without the slow-fast split performed substantially worse. All variants retained matched task-specific training configurations. Rotation contributed 0.65–3.53 points, with the largest effect on the long

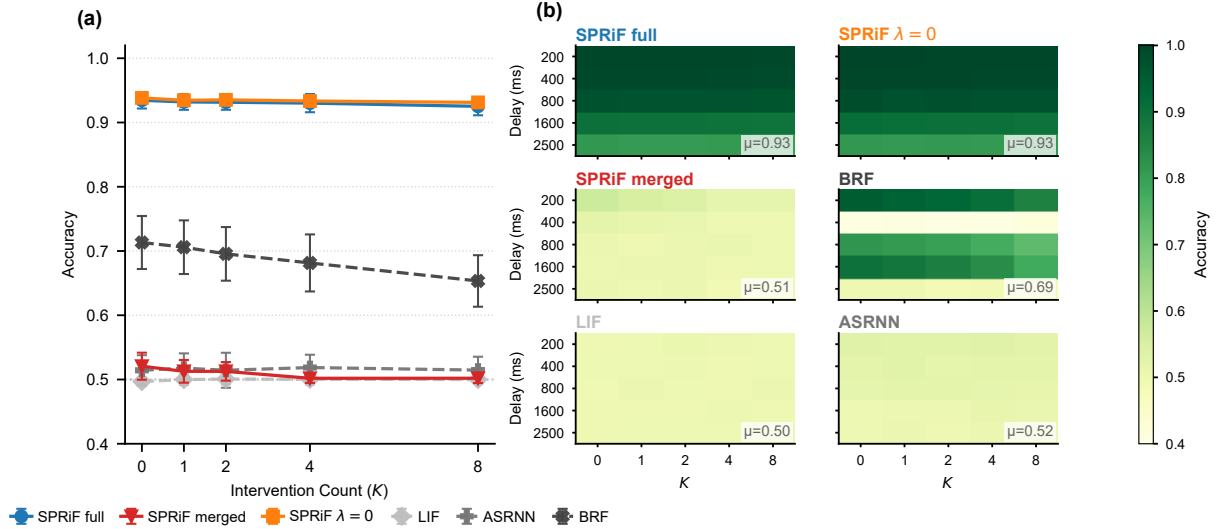


Figure 1: Controlled spike-intervention results (SI-DMS). (a) Accuracy versus intervention count K for the three SPRiF mechanism variants (solid, saturated) and three external baselines (dashed, gray); values are 3-seed means $\pm$ seed standard deviations. (b) Per-model delay-by- K accuracy grids (3-seed mean) for all six models with a shared colorbar; μ denotes the per-model mean across the grid.

Table 2: Mechanism ablations (accuracy, %). Each component contributes on all three datasets, with merging memory and discharge causing the largest degradation.

Dataset	Full	$\omega=0$	Merged	$\lambda=0$
PS-MNIST	95.82	92.29	93.44	95.32
GSC	94.55	93.83	90.05	94.22
QTDB	88.43	87.78	83.39	87.40

permuted sequence. Projective reset provided a smaller but consistent gain of 0.33–1.03 points. Across these settings, state separation was the largest ablated component, whereas spectral rotation and a learnable reset direction further refined temporal processing.

4.4 Controlled Spike Intervention

To test whether slow-fast separation protects memory under reset stress, we designed a Spike-Intervention Delayed Match-to-Sample (SI-DMS) task. A cue was followed by a delay, during which selected hidden neurons were forced to spike at controlled times. A match/non-match probe then tested recall. At each intervention timestep, 10% of hidden neurons were forced to spike before thresholding, so that the standard reset mechanism was applied. We evaluated $K \in \{0, 1, 2, 4, 8\}$ intervention times. The maximum-stress setting, $K = 8$, was not used during training. If reset reaches the memory-bearing state, these interventions should impair delay-period recall. If the slow state is structurally insulated, recall should remain more stable.

Table 3 reports clean ($K = 0$) and maximum-stress ($K = 8$) accuracy averaged over three seeds. The fully

Table 3: SI-DMS results (3-seed average). Clean and maximum-stress accuracies correspond to $K = 0$ and $K = 8$, respectively; “Drop” is the stress-induced accuracy loss. External baselines are marked (ext.). The separated SPRiF variants maintain high accuracy under intervention.

Model	Clean Acc.	Stress Acc.	Drop
sprif_full	0.928	0.921	0.007
sprif_lambda0	0.935	0.929	0.006
sprif_merged	0.517	0.504	0.013
LIF (ext.)	0.500	0.502	−0.002
ASRNN (ext.)	0.520	0.517	0.003
BRF (ext.)	0.704	0.649	0.055

separated variant, sprif_full, reached 92.8% clean accuracy and lost 0.7 points under intervention stress. Removing the projective reset direction ($\lambda=0$; sprif_lambda0) produced a comparable 0.6-point drop. This pattern indicates that structural separation, rather than λ alone, was important under this protocol. The merged variant, sprif_merged, performed near chance in both conditions (51.7% clean and 50.4% under maximum stress). Among the external baselines, BRF decreased from 70.4% to 64.9%, whereas LIF and ASRNN remained near chance throughout. Under matched SI-DMS controls, these results support the claim that functional state decomposition protects delay-period memory from spike-triggered disruption. Figure 1 shows the delay-by-intervention grid. Complete grids and integrity audits are provided in the appendix.

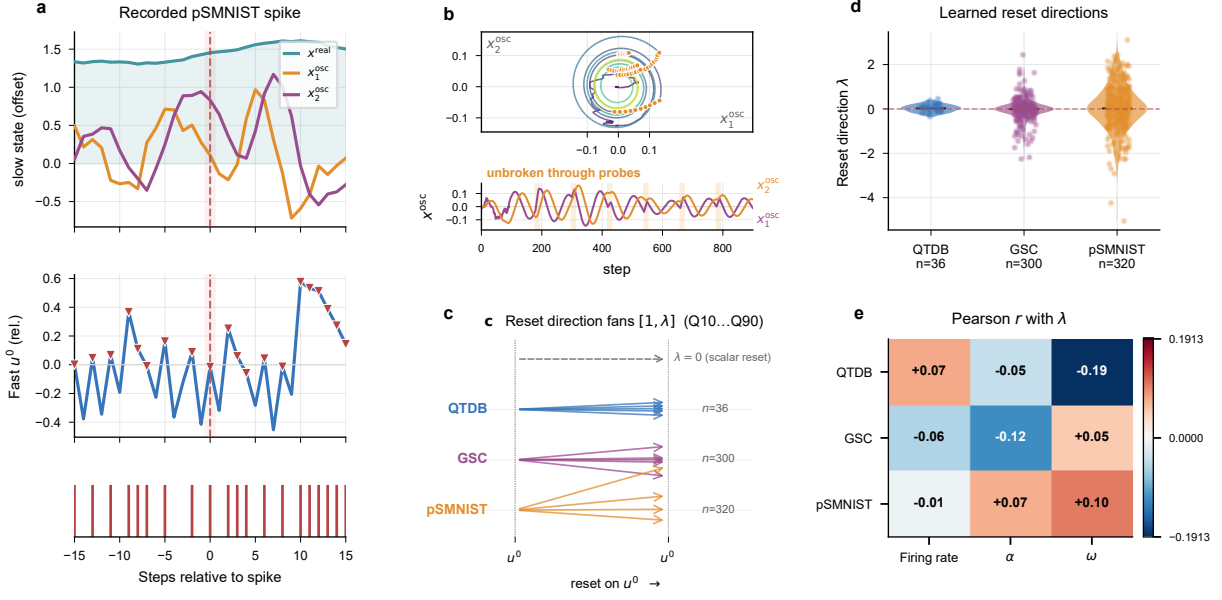


Figure 2: Functional decomposition and learned projective reset. (a) Slow and fast states of a trained PS-MNIST neuron around a recorded spike; slow coordinates evolve continuously while the fast membrane trace changes abruptly. (b) Oscillatory slow-state trajectory in a cue-delay task: the 2D phase plane (top) marks probe times, and the x_1^{osc} and x_2^{osc} time series (bottom) traverse the orange probe windows without a discontinuity, illustrating continuous slow-state evolution rather than an event-level reset test. (c) Representative learned reset directions $[1, \lambda]$ from task-wise quantiles. (d) Full distributions of λ . (e) Pearson correlations between λ and firing rate, α , and ω .

4.5 Dynamical Analysis

The update equations ensure that reset does not directly modify x_t . SI-DMS tests the functional consequence of this design under controlled interventions. Figure 2a shows the resulting state dynamics around recorded spikes. Figure 2b shows a slow-state oscillation across probe windows in a cue-delay task. It illustrates continuous slow-state evolution, rather than providing an event-level reset test. Figure 2c–e examines the learned projective reset directions.

Projective reset did not collapse to a shared direction. Across QTDB, GSC, and PS-MNIST, the learned λ values spanned both signs (Fig. 2c–d). Their correlations with firing rate, α , and ω were weak, with $|r| \leq 0.191$ (Fig. 2e). Together with the $\lambda = 0$ ablation loss, these observations support diverse reset directions in the fast state. They do not assign a universal biological meaning to either sign.

We next injected a unit impulse into trained neurons and recorded 100 steps of intrinsic slow-state evolution. Figure 3 visualizes the resulting temporal basis. α -stratified examples span rapidly decaying and persistent real traces, whereas the oscillatory coordinates vary in sign and frequency. All-neuron spectra show broad within-task variation in effective low-pass behaviour. In selected slow-decay examples, the SPRiF real modes had derived time constants of approximately 272 steps on GSC and 138 on QTDB. The closest available trained ASRNN membrane constants were 34 and 46 steps. This comparison is limited to GSC and QTDB, the only two tasks with trained ASRNN kernels in the saved anal-

ysis data. Additional impulse-response and spectral analyses are provided in the appendix.

5 Limitations

SPRiF replaces one scalar state with five state values per neuron. It therefore increases activation memory and neuron-local arithmetic, even when the end-to-end parameter count is small. Several baselines are drawn from their original papers under task-specific protocols. Unified reimplementations would be needed for fully controlled significance claims. The current evaluation focuses on small- to medium-scale temporal classification and does not establish energy gains on neuromorphic hardware. The diagnostic analyses demonstrate robustness under the SI-DMS intervention protocol and reveal diverse kernels, but do not establish that every learned mode is necessary for every input. These boundaries motivate matched wall-clock and spike-operation measurements, together with larger long-context benchmarks.

6 Conclusion

SPRiF is built on a simple principle: the state that stores temporal context need not be reset after a spike. A constrained spectral slow state supplies real and rotational memory modes. A fast discharge state produces binary spikes and receives a learnable projective reset. Across five temporal benchmarks, this design attained the highest mean accuracy in our comparison on three tasks and remained competitive on the other two with modest network size. Ablations identi-

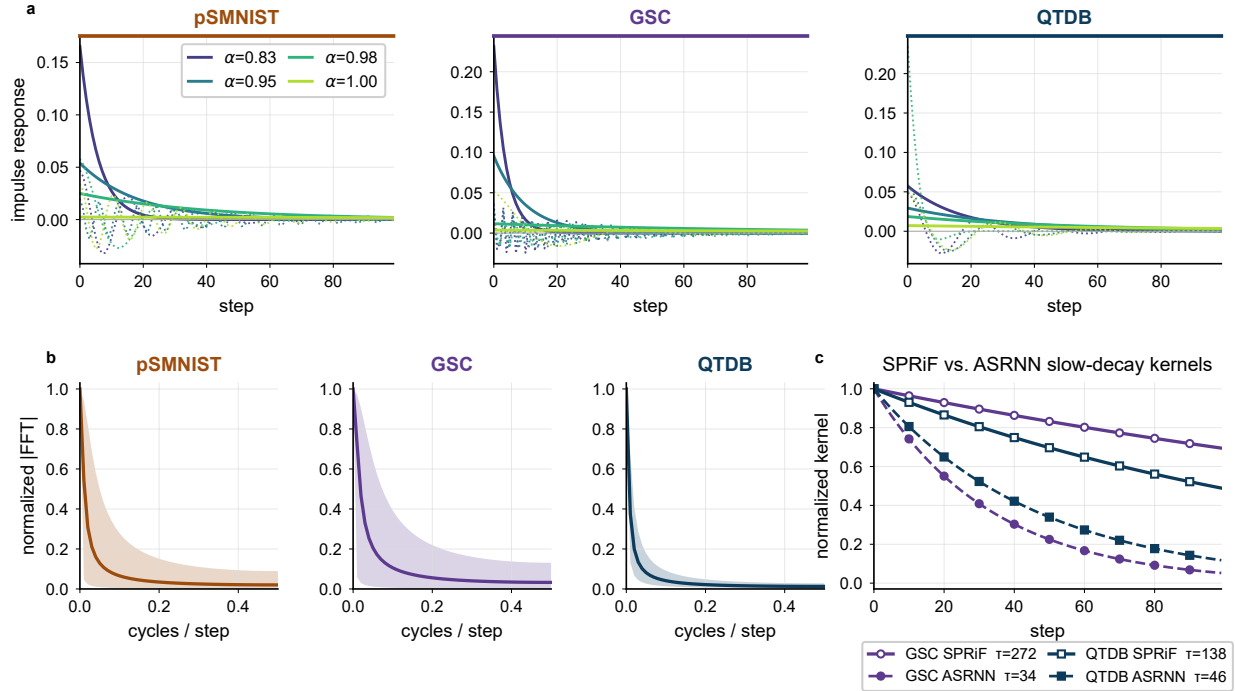


Figure 3: Learned temporal kernels from impulse-response analysis. (a) Real (solid) and oscillatory (dotted) slow-state coordinates of four α -quantile SPRiF neurons per task over 100 post-impulse timesteps; traces span rapidly decaying to persistent regimes. (b) Median normalized real-coordinate amplitude spectrum with the 10–90% range aggregated over all neurons, per task. (c) Representative slow-memory SPRiF kernels (real coordinate) and the closest available trained ASRNN membrane kernels on GSC and QTDB; derived time constants τ (timesteps) are shown.

fied slow–fast separation as the largest contributor, with additional gains from rotation and projective reset. Controlled spike-intervention and impulse-response analyses provided complementary evidence for the proposed mechanism. These findings support functional state decomposition as a reusable neuron-level design direction. Larger-scale evaluation and neuromorphic-hardware deployment remain open.

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